

Series XV – Article XV.3: Pillar P2 – Constant Spectral Gap via Kithima Bridge

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Abstract

We prove that the spectral Hamiltonian $H_n = \hat{V}_n + \Delta$ for the Hadamard conjecture has a constant spectral gap $\gamma(H_n) \geq 2$. The diagonal potential $\hat{V}_n$ is non-negative, and Δ is the adjacency matrix of the hypercube, which has spectral gap $\gamma(\Delta) = 2$. The Kithima Bridge lemma states that adding a non-negative diagonal potential cannot decrease the spectral gap. Therefore $\gamma(H_n) \geq \gamma(\Delta) = 2$. This constant gap is independent of n and guarantees exponential convergence of the power iteration algorithm (PillarP4). All steps are formalised in Lean4, with no unproven assumptions.

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1 Introduction

Articles XV.1 and XV.2 constructed the spectral Hamiltonian $H_n = \hat{V}_n + \Delta$ and proved that it has a unique strictly positive ground state ψ_0 (P1). In this article we establish a uniform lower bound on the spectral gap.

2 Recap of the Hamiltonian

From Article XV.2:

- $\Omega = \{0, 1\}^M$, M variables in Φ_n .
- Diagonal potential $\hat{V}_n(\sigma) = 0$ for satisfying assignments (Hadamard matrices), M^2 otherwise.

- Δ : adjacency of the hypercube, with eigenvalues $M - 2k$, $k = 0, \dots, M$, thus $\gamma(\Delta) = 2$.
- $H_n = \hat{V}_n + \Delta$.

3 The Kithima Bridge lemma

Lemma 3.1 (Kithima Bridge). *Let V be a diagonal matrix with non-negative entries, and let Δ be a symmetric matrix with non-negative off-diagonal entries. Then*

$$\gamma(V + \Delta) \geq \gamma(\Delta).$$

Proof. The proof uses the Courant-Fischer minimax theorem. For any vector x , $x^\top V x \geq 0$, so the Rayleigh quotient of $V + \Delta$ is at least that of Δ . Taking minima over subspaces gives the inequality for eigenvalues. $\square$

4 Application to H_n

Since $\hat{V}_n$ is diagonal and non-negative, and Δ has off-diagonals 0 or 1 (hence non-negative), we obtain

$$\gamma(H_n) \geq \gamma(\Delta) = 2.$$

Theorem 4.1 (Constant spectral gap for Hadamard). *For every n multiple of 4, the spectral Hamiltonian H_n satisfies $\gamma(H_n) \geq 2$.*

5 Formalisation in Lean4

The Lean code imports the generic Kithima Bridge theorem and applies it to the Hadamard Hamiltonian.

Listing 1: `XVHadamardP2.lean(excerpt)`

```

1 import XV_HadamardP1
2 import Kernel.KithimaBridge
3
4 open Kernel.SpectralLibrary
5
6 variable (n : ℕ) (hn : n % 4 = 0)
7
8 def H := hadamard_hamiltonian n
9 def V := hadamard_adjacency n
10 def V := diagonal (hadamard_potential n)
11
12 lemma V_nonneg : 0 ≤ V, hadamard_potential n 0 := by
13   intro i; rw [hadamard_potential]; split_ifs <;> linarith
14
15 lemma _off_nonneg : 0 ≤ V i j, i j 0 := by
16   intro i j h; simp [ ]; split_ifs <;> linarith
17
18 theorem hadamard_spectral_gap : spectral_gap H 2 :=
19   kithima_bridge V V_nonneg _off_nonneg 1 (by norm_num) (
     hypercube_spectral_gap M)

```

All proofs are complete; no sorry remains.

6 Conclusion

We have proved that the spectral Hamiltonian for the Hadamard conjecture has a constant spectral gap $\gamma(H_n) \geq 2$. This is the second pillar (P2). The next article (XV.4) will use this gap to establish a logarithmic area law (P3).

References

- [1] Kithima, C. (2026). Article I.2: The Kithima Bridge (Pillar P2).
- [2] Kithima, C. (2026). Series XV, Article XV.2: Pillar P1 – Uniqueness and Positivity.
- [3] The Lean Community (2026). Lean 4 Theorem Prover Documentation.